

Dundee peripheral blood mononuclear cell (PBMC) isolation protocol (from whole blood)

Authors: Adil R. Sarhan¹, Andrew J.M. Howden², Dario R. Alessi¹ and Esther M. Sammler¹

1 Medical Research Council Protein Phosphorylation and Ubiquitylation Unit, School of Life Sciences, University of Dundee, Dow Street, Dundee DD1 5EH, UK

2. Division of Cell Signaling and Immunology, School of Life Sciences, University of Dundee, Dow Street, Dundee DD1 5EH, UK

1. MATERIALS

1.1. Consumables:

- 1.1.1. SepMate™ tube (STEMCELL, Cat# 85450), 50 mL capacity.
- 1.1.2. BD Vacutainer sodium heparin 6ml tubes, green BD Hemogard Closure 13x100mm (BD, cat# 367876). (3 tubes are needed for each PBMC preparation).
- 1.1.3. 20 ml syringes (BD, cat# 300613)
- 1.1.4. 21-G needle (for example BD, # 304432)
- 1.1.5. 50 mL falcon tube (CELLSTAR, Cat #227 261).
- 1.1.6. 15 mL falcon tube (CELLSTAR, Cat #188 271).
- 1.1.7. 1.5 mL Eppendorf tubes.
- 1.1.8. Marker pen.
- 1.1.9. Ice.
- 1.1.10. Liquid nitrogen.
- 1.1.11. Dry Ice and large polystyrene box for shipping
- 1.1.12. Ethanol, in spray bottle.
- 1.1.13. Pipette tips (serological 10 mL, 25 mL & 1 mL, 200 µL, 10 µL).
- 1.1.14. Personal protection equipment:
 - 1.1.14.1. Disposable gloves.

- 1.1.14.2. Lab coat.
- 1.1.14.3. Safety glasses

1.2. Reagents:

- 1.2.1. Ficoll-Paque PREMIUM (GE Healthcare Cat# 17-5442-02).
- 1.2.2. Dulbecco's phosphate-buffered saline (PBS) (ThermoFisher Cat# 14190094) with 2% Fetal bovine serum (FBS), (Life Sciences Production, Cat # S-001-BR). Prepare fresh on day of use. 50ml of PBS with 2% FBS are needed per PBMC preparation: add 1ml of FBS to 49ml of PBS. Use at room temperature.
- 1.2.1.1. MLi-2 (LRRK2 inhibitor diluted in DMSO at a stock concentration of 200 μ M (1000X concentration), available from MRC-PPU Reagents to order please email: MRCPPUreagents@dundee.ac.uk.
- 1.2.3. DMSO (Sigma, Cat #D8418).
- 1.2.4. DIFP (Diisopropylfluorophosphate) (Sigma, Cat #D0879). Prepare 0.5M stock solution in isopropanol using special precautions (please see 1.2.6.)
- 1.2.5. Isopropanol (anhydrous grade) (Sigma, Cat #278475)
- 1.2.6. Lysis buffer (50 mM Tris-HCl pH 7.5, 1%(v/v) Triton X-100, 1 mM EGTA, 1 mM Na₃VO₄, 50 mM NaF, 10 mM β -glycerophosphate, 5 mM sodium pyrophosphate, 1 μ g/ml Microcystin-LR, 0.27 M sucrose, 0.1% (v/v) β -mercaptoethanol, 1x cOmplete(EDTA-free) protease inhibitor cocktail (Roche) + 1 mM diisopropylfluorophosphate (DIFP).
 - 1.2.6.1.Note: Diisopropylfluorophosphate (DIFP) is prepared as a 0.5 M stock solution in isopropanol (it is unstable in aqueous solution) and stored at - 80 °C. Please note that DIFP is toxic and should be handled with care in the fume hood. DIFP can be added to the lysis buffer and used immediately. Alternatively, complete lysis buffer containing DIFP and all other components can be aliquoted and stored at - 80 °C for subsequent use for at least 4 weeks.
 - 1.2.6.2.Note we purchase Microcystin-LR from Enzo Life Sciences, Cat# number ALX-350-012-M001) and make 1 mg/ml stocks in DMSO and store at -80 °C.
 - 1.2.6.3.We can provide frozen lysis buffer in aliquots without Microcystin-LR and DIFP; to order please email MRCPPUreagents@dundee.ac.uk.

- 1.3. Equipment:**
- 1.3.1. Category 2 biological safety cabinet.
 - 1.3.2. Centrifuges (Beckman Coulter Allegra X-15R & Eppendorf centrifuge 5417R) or equivalent. Swinging Bucket Rotor for 15 mL and 50 mL falcon tubes at speed of 1000-1200 xg.
 - 1.3.3. Pipette tips and pipettes.
 - 1.3.4. Sharps waste container.
 - 1.3.5. Freezer, -80 °C.
 - 1.3.6. Liquid nitrogen carrier

2. METHOD

According to local UK regulation, we undertake all manipulations and pipetting of human blood in a category 2 biological safety cabinet.

- 2.1. Collect 3 x 6 ml of blood into BD Vacutainer sodium heparin 6ml tubes, green BD Hemogard Closure 13x100mm (BD, cat# 367876). Mix gently by inverting tubes 8 - 10 times.
- 2.2. Add density gradient medium Ficoll-Paque PREMIUM to the SepMate™ tube (STEMCELL, Cat# 85450, 50 mL capacity) using a 20ml syringe with a large bore needle (at least 21G) by carefully adding the 15 mL through the central hole of the SepMate™ insert. Dispose of needle appropriately. The top of the density gradient medium will be above the insert. NOTE: Small bubbles may be present in the density gradient medium after pipetting. These bubbles will not affect performance.
- 2.3. Using a 5 ml pipette, transfer 15 ml of blood from the BD Sodium heparin tubes into a 50 ml Falcon tube and dilute at a 1:1 ratio by adding 15 ml of (PBS with 2% FBS). Mix gently by inversion.

- 2.4. Keeping the SepMate™ tube vertical, add the diluted blood sample by pipetting it down the side of the tube. Take care not to pour the diluted blood sample directly through the central hole that lies in the centre of the insert disk.
- 2.5. Centrifuge at 1200 x g for 10 minutes at room temperature, with the brake ON.
- 2.6. After centrifugation, carefully pipette off the top plasma transparent layer and discard appropriately.
- 2.7. Using a new pipette, collect the remainder of the sample above the insert disk (which contains the PBMCs) into a 15 mL falcon tube.
- 2.8. Top up the tube to 15 mL of (PBS with 2% FBS). Mix gently by inversion.
- 2.9. Centrifuge at 1000 x g for 2 minutes, brake ON, to pellet the PBMCs.
- 2.10. Carefully discard the supernatant without disturbing the PBMC pellet.
- 2.11. Carefully resuspend the cell pellet in 10 mL of (PBS with 2% FBS)
- 2.12. Centrifuge at 1000 x g for 2 minutes, brake ON to pellet the PBMCs.
- 2.13. During the centrifugation Step 2.12, take MLI-2 inhibitor stock solution (200 µM / 1000x) out of the -80 freezer, and leave at room temperature to thaw for subsequent use (Steps 2.19 and 2.21)
- 2.14. Immediately after the centrifugation, carefully discard the supernatant without disturbing the PBMC pellet.
- 2.15. Resuspend the PBMC cell pellet in 10 mL (PBS with 2% FBS) by gently pipetting cells up and down 4 times.
- 2.16. Divide resuspended PBMCs equally into 2 tubes (5 ml in each).
- 2.17. Label one tube “DMSO” and other tube “MLi-2”.
- 2.18. To “DMSO” labelled tube, add 5 µL of DMSO and mix gently by pipetting up and down 2-times with a 10 ml pipette.
- 2.19. To “MLi-2” labelled tube, add 5 µL of 200 µM MLI-2 stock solution (final concentration 200 nM) and mix gently by pipetting up and down 2-times with a 10 ml pipette.
- 2.20. Incubate samples for 30 minutes at room temperature. Mix gently by inversion every 10 minutes during the incubation period. Incubate samples for 30 minutes at room temperature. Mix gently by inversion every 10 minutes during the incubation.
- 2.21. During the incubation period,
 - remove 0.5 M DIFP stock from -80 °C freezer and place in fume hood on ice.

- remove 1 mg/ml Microcystin-LR stock from -80 °C freezer and place at room temperature to thaw.
 - defrost an aliquot (0.5 ml) of the lysis buffer by taking it out of the freezer allow to defrost at room temperature and then place it on ice for subsequent use (Step 2.27).
 - prepare 1 mL of (PBS with 2% FBS) containing 1 µL of DMSO and call this “DMSO Resuspension Buffer”
 - prepare 1 mL of (PBS with 2% FBS) containing 1 µL of 200 µM MLi-2 stock solution and call this “MLi-2 Resuspension Buffer”
- 2.22. After the 30 minutes incubation, centrifuge both tubes at 335 x g for 5 mins. The setting for acceleration and deceleration is both 5 using a Beckman Coulter Allegra X-15R Centrifuge (maximum is 10) to pellet the PBMCs.
- 2.23. Carefully discard the supernatant in each tube without disturbing the PBMC pellets.
- 2.24. For the DMSO labelled sample gently resuspend pellet in 1 ml of “DMSO resuspension buffer” and transfer to Eppendorf tube labelled “DMSO”.
- 2.25. For the MLi-2 labelled sample gently resuspend pellet in 1 ml of “MLi-2 resuspension buffer” and transfer to Eppendorf tube labelled “MLi-2”.
- 2.26. Centrifuge both tubes at 300 x g for 3 min.
- 2.27. During this time proceed with preparing the lysis buffer: In the fume hood carefully to the 0.5 ml lysis buffer add 0.5 µl of 0.5 M DIFP solution as well as 0.5 µl of 1 mg/ml microcystin-LR. Mix and leave on ice until use. DIFP should be added to lysis buffer within 15 min of cell lysis (Steps 2.31 and 2.32) as DIFP relatively unstable in aqueous solution.
- 2.28. Return 0.5M DIFP to -80 °C freezer.
- 2.29. Immediately after the centrifugation in Step 2.26, carefully discard supernatant by using a pipette while taking care not to disturb the PBMC pellets.
- 2.30. Place the tubes with the PBMC cell pellets on ice.
- 2.31. Immediately add 200 µL of lysis buffer containing DIFP to the “DMSO” labelled tube. Using a 100-200 µL pipette, re-suspend the cell pellet by pipetting up and down until all cells are lysed (5-10 times).
- 2.32. Next add 200 µL of lysis buffer containing DIFP to the “DMSO” labelled tube. Using a 100-200 µL pipette, re-suspend the cell pellet by pipetting up and down until all cells are lysed (5-10 times).

- 2.33. Incubate cell lysates on ice for 10 mins. Place “DMSO” and “MLi-2” labelled tubes into a centrifuge and remove cell debris by centrifugation at 14,000 rpm for 15 mins at 4 °C.
- 2.34. Transfer ”DMSO” and “MLi-2” supernatants (containing PBMC lysates) into new Eppendorf tubes. Discard debris pellet.
- 2.35. Immediately snap freeze samples in liquid nitrogen.
- 2.36. Store samples at -80 °C until further use or shipment on dry ice keeping samples frozen at all times.